CHAPTER

3

ELASTICITY OF DEMAND

Learning Objectives

The students will learn:

- *Meaning of elasticity of demand and various types of elasticity of demand.*
- *The various degrees of elasticity of demand.*
- *The various methods for the calculation of elasticity of demand.*
- *The factors that affect the elasticity of demand.*

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 45

Revision Notes

➢ **Elasticity of Demand:** The concept of elasticity of demand, refers to the degree of responsiveness of quantity demanded of a goods to a change in the price of the commodity, income of the consumer and the prices of the related goods, that is substitutes and complements.

- Mathematically, it is represented as

$$\text{Elasticity of Demand} = \frac{\text{Percentage change in the quantity Demanded of the commodity}}{\text{Percentage Change in the Price of the commodity}} \quad \text{Ed} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Scan to know more about this topic
Elasticity of Demand

➢ **Types of Elasticity of Demand:**

(i) Price Elasticity of Demand: It is the degree of responsiveness of the quantity demanded of a commodity with respect to the change in the price.

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

(ii) Income Elasticity of Demand: Income elasticity of demand represents the degree of responsiveness of the quantity of a particular commodity demanded by an individual when there is a change in his income.

$$\text{Elasticity of Demand} = \frac{\text{Percentage change in the quantity demanded of the commodity}}{\text{Percentage change in the income of the consumer}}$$

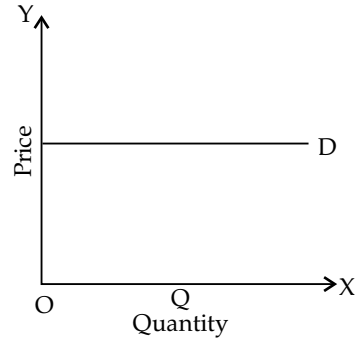
Scan to know more about this topic
Degrees of Price elasticity of demand.

(iii) Cross Elasticity of Demand: It is related to the concept of cross demand. Cross Elasticity of Demand is the responsiveness of the change in the demand of a commodity to the change in the price of the substitute good or complement good.

$$\text{Elasticity of Demand} = \frac{\text{Percentage change in the quantity demanded of the commodity X}}{\text{Percentage change in the Price of the commodity Y}}$$

➢ **Different Magnitudes of price Elasticity of Demand:**

(i) Perfectly elastic demand ($E_d = \infty$): When a small change in price of a product causes a major change in its demand, it is said to be perfectly elastic demand. It is shown by a horizontal straight line parallel to X-axis.



Scan to know more about this topic
Factors affecting Price Elasticity of Demand

Elasticity of Demand

Types

Price elasticity of Demand

Is the measure of degrees of responsiveness of demand for a commodity to change in its price.

$$e_d = \frac{\text{\% Change in demand}}{\text{\% Change in price}}$$

Income Elasticity of Demand

Is the measure of the degree of change in demand in response to change in income of the consumers.

$$e_d = \frac{\text{\% Change in demand}}{\text{\% Change in income}}$$

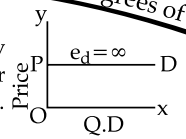
Cross Price Elasticity of Demand

Is the measure of a change in the demand for one product in response to a change in the price of other product.

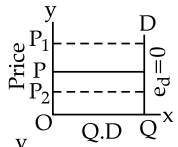
$$e_d = \frac{\text{\% Change in } D_x}{\text{\% Change in } P_y}$$

Degrees of Elasticity of Demand

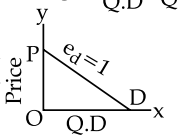
1. Perfectly elastic demand ($e_d = \infty$)
   When demand changes to any extent without any change or with very little change in its price.



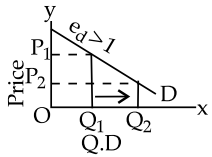
2. Perfectly inelastic demand ($e_d = 0$)
   When quantity demanded does not change at all with any change in its price.



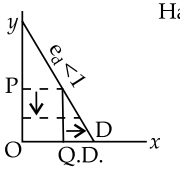
3. Unitary elastic demand ($e_d = 1$)
   When percentage change in demand for a commodity is equal to percentage change in its price.



4. Highly elastic demand ($e_d > 1$)
   When percentage or proportionate change demand for a commodity is more than the percentage or proportionate change in its price.



5. Highly inelastic demand ($e_d < 1$)
   When percentage or proportionate change in demand for a commodity is less than the percentage or proportionate change in its price.



Methods of measuring e_d of Demand

Percentage or proportionate method

$$e_d = \frac{\text{\% change in quantity demanded of the commodity}}{\text{\% change in the price of the commodity}}$$

or $e_d = \frac{\Delta Q}{\Delta P} \cdot \frac{Q}{P}$, or $e_d = \frac{\Delta Q}{\Delta P} \cdot \left(\frac{Q}{P}\right)$

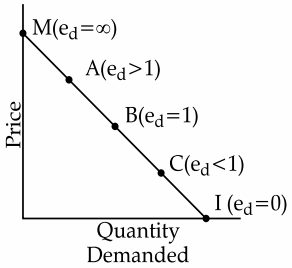
ΔQ = Change in quantity demanded
ΔP = Change in price of demanded
P = Original price
Q = Original or initial demand of a Commodity

Total Outlay or Expenditure Method

- Indicates the direction in which total expenditure (TE) on a product changes as a result of change into price (P)
  - P↑TE↓ $= e_d > 1$
  - P↓TE↑ $= e_d < 1$
  - P↑↓TE $= e_d = 1$

Point Method or Geometric Method

$$e_d = \frac{\text{Lower sector of demand curve}}{\text{Upper sector of demand curve}}$$



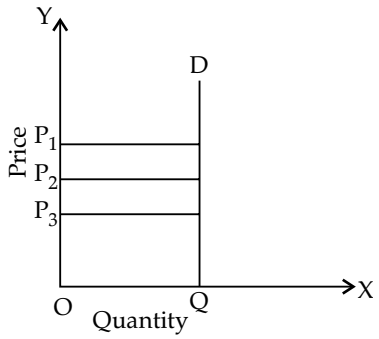
Factors affecting elasticity of Demand

- Nature of Commodity
  - Luxuries $e_d > 1$
  - necessities $e_d < 1$
- Availability of Substitutes
  - More $e_d > 1$
  - less $e_d < 1$
- Share in total Expenditure
  - More $e_d > 1$
  - less $e_d < 1$
- Different Uses of Commodity
  - More $e_d > 1$
  - less $e_d < 1$
- Possibility to Postpone the Consumption
  - Can be $e_d > 1$
  - Can't be $e_d < 1$
- Income of the Consumer
  - Lower, $e_d > 1$
  - Higher, $e_d < 1$
- Price level
  - Middle, $e_d > 1$
  - High/Low, $e_d < 1$
- Time period
  - Long run, $e_d > 1$
  - Short run, $e_d < 1$
- Consumer habits
  - Inhabitual $e_d > 1$
  - Habitual $e_d < 1$

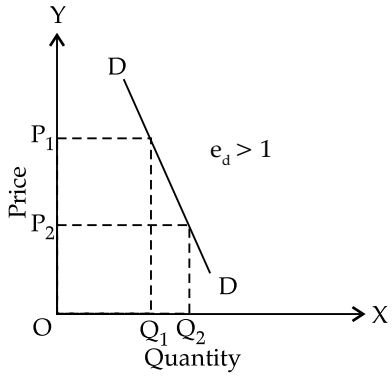
Trace the Mind Map

▸ First Level ▸ Second Level ▸ Third Level

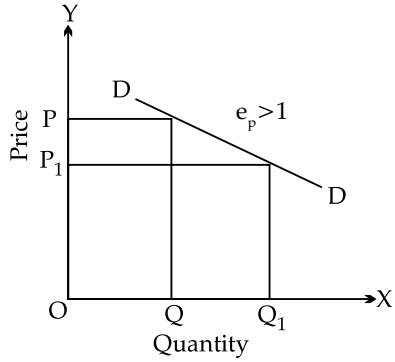
(ii) Perfectly inelastic demand ($E_d = 0$): A perfectly inelastic demand is one when there is no change produced in the demand of a product with change in its price. It is shown by a vertical straight line parallel to Y axis.



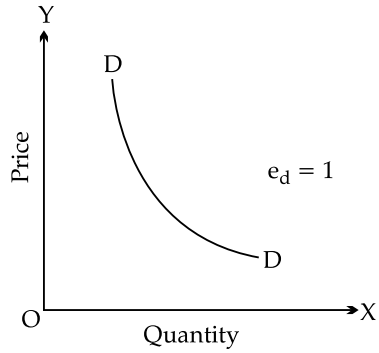
(iii) Relatively Inelastic demand ($E_d < 1$): Relatively inelastic demand is one when the percentage change produced in demand is less than the percentage change in the price of a product.



(iv) Relatively Elastic demand ($E_d < 1$): Relatively high elastic demand refers to the demand when the proportionate change produced in demand is greater than the proportionate change in price of a product.



(v) Unitary elastic demand ($E_d = 1$): When the proportionate change in demand brings the same change in the price of the product, the demand is referred as unitary elastic demand.



- **Percentage Method of Measuring of Elasticity of Demand:** The percentage method measures the price elasticity of demand as the percentage change in the quantity demanded to the percentage change in the price of the commodity.
- It is measured using the following formula:

$$\text{Elasticity of Demand} = \frac{\text{Percentage change in the quantity demanded of the commodity}}{\text{Percentage change in the price of the commodity}}$$

$$E_d = \frac{\frac{\Delta Q}{Q}\times 100}{\frac{\Delta P}{P}\times 100} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P}\times\frac{P}{Q}$$

- **Arc Elasticity Method:** Arc elasticity of demand measures elasticity between two points on a curve – using a mid-point between the two curves.

  Calculating Arc Elasticity of Demand

$$\text{Midpoint } Q = \frac{(Q1 + Q2)}{2}$$

$$\text{Midpoint } P = \frac{(P1 + P2)}{2}$$

  Once we have the midpoint, we calculate the PED in the usual way

$$\text{Arc elasticity of demand (PED)} = \frac{Q2 - Q1 / \text{midpoint } Q}{P2 - P1 / \text{midpoint } P}$$

- **Total Expenditure Method:** Under this method, impact (effect) of change in price on the expenditure of a goods is studied. When price of a goods changes, consumer's total expenditure on it may increase, decrease, or remain constant. Thus, elasticity is measured by comparing the total expenditure made on the goods before and after the price change.
- **Geometric or Point Method:** In order to measure Ed at any particular point, lower portion of the curve from that point is divided by the upper portion of the curve from the same point.

  Formula: Elasticity of Demand (E_d)

$$= \frac{\text{Lower segment of demand curve (LS)}}{\text{Upper segment of demand curve (US)}}$$

➢ **Factors Affecting Price Elasticity of Demand**

- **Objective Factors:** (1) Nature of commodity, (2) Existence of substitutes, (3) Alternative uses of a commodity, (4) Postponement of consumption, and (5) Joint demand.
- **Subjective Factors:** (1) Habits of consumers, (2) Change in income of consumers, (3) Standard of living of people, (4) Share in total expenditure, and (5) Class of buyers.
- **Social Factors:** (1) Distribution of National Income, and (2) Rationing System.
- **Price Factors:** (1) General price level, and (2) Effect of time element.

Examples:

1. **The price of a commodity falls from ₹5 to ₹3, resulting an increase in the purchase of the commodity from 20 units to 22 units. Calculate the Price Elasticity of Demand.**

Ans. Here, $P_1 = ₹5, Q_1 = 20, P_2 = ₹3, Q_2 = 22$

$\Delta P = 3 - 5 = -2,$

$\Delta Q = 22 - 20 = 2$

We have to find Price Elasticity of Demand

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$= \frac{2}{-2} \times \frac{5}{20}$$

$$= 0.25$$

Therefore, Price Elasticity of Demand = 0.25

2. **Price Elasticity of Demand of a product is unity. A household buys 100 units of this product when its price is ₹20 per unit. If its price rises to ₹24 per unit, how much quantity of the product will be bought by the household?**

Ans. Here, $P_1 = ₹20,$

$Q_1 = 100,$

$P_2 = ₹24,$

$Q_2 = ?$

$E_d = 1$

$\Delta P = 24 - 20 = 4$

$\Delta Q = Q_2 - 100$

$$Ed = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$-1 = \frac{Q_2 - 100}{4} \times \frac{20}{100}$$

$$-20 = Q_2 - 100$$

$$Q_2 = 80 \text{ units}$$

Hence, the new quantity demanded is 80 units.

Key Terms

➢ **Demand:** The desire to purchase a commodity backed by willingness to purchase and the capacity to purchase it at a given time.

➢ **Cross Demand:** Demand of the commodity with respect to the price of other related commodity.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Define elasticity of demand. [OEB]

Ans. Elasticity of demand refers to the degree of responsiveness of quantity demanded of a goods to a change in the price of the commodity, income of the consumer and the prices of the related goods, that is substitutes and complements.

Q.2. Suppose the price elasticity of demand for a good is – 0.2. How will the expenditure on the good be affected if there is a 10% increase in the price of the good? [OEB]

Ans. Total expenditure will rise if there is a 10% rise in the price of the good, since its demand is inelastic.

Q.3. If two demand curves intersect, which one has the higher price elasticity? AI [OEB]

Ans. When two demand curves intersect, the flatter curve is more elastic.

Q.4. Give difference between elastic demand and inelastic demand? [OEB]

Ans. Elastic demand can be defined as when percentage change in demand for a good is greater than percentage change in price of good. Inelastic demand can be defined when the percentage change in the demand for a good is less than percentage change in the price of the good.

Q.5. Give the meaning of percentage method of measuring price elasticity of demand. [OEB]

Ans. The percentage change in quantity demanded to the percentage change in the price of the commodity is the price elasticity of demand.

Q.6. State whether the given statement is true of false with reason: [OEB]

(a) "When due to percentage change in price of substitute goods there is percentage change in demand for the given good, then it is called as income elasticity of demand.

Ans. False.

Reason: When due to percentage change in price of substitute goods there is percentage change in demand for the given good, then it is called as cross elasticity of demand.

(b) "According to expenditure method, elasticity of demand is less than one when price and total expenditure have inverse relationship." [OEB]

Ans. False.

Reason: According to expenditure method, when price and total expenditure move in the same direction then elasticity of demand is less than one. That means when prices rise, total expenditure also rises, and when price falls, total expenditure also falls.

(c) "The price elasticity of demand for goods which has many uses is higher." [OEB]

Ans. True.

Reason: The greater number of uses of a commodity, the higher is the price elasticity of demand. Example, milk can be used to make cheese, butter, curd, etc. If its price rises, it will be put to only important uses like serving the children or for the sick members in the family.

(d) "When the percentage change in demand for a good is equal to percentage change in price of a good, then it is called as less than unit elastic demand." [OEB]

Ans. False.

Reason: When the percentage change in demand for a good is equal to percentage change in price of a good, then it is called as unitary elastic demand.

2 Marks Questions

Q.1. The market demand for a commodity at ₹8 per unit is 100 units. The price rises and as a result, its market demand falls to 75 units. Find out the new price of the commodity if the price elasticity of demand is -1. [ISC Specimen Paper, 2023-24]

Concept Applied: Price Elasticity of Demand

Ans.

$P_1 = ₹8$

$P_2 = ?$

$Q_1 = 100$ units

$Q_2 = 75$ units

$E_d = -1$

$\Delta P = P_2 - 8$

$\Delta Q = 75 - 100 = -25$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$-1 = \frac{-25}{P_2 - 8} \times \frac{8}{100}$$

$$1 = \frac{1}{P_2 - 8} \times 2$$

$P_2 - 8 = 2$

$P_2 = 10$

Hence, the new price is ₹10

Q.2. The price of a goods decreases from 100 to 80 per unit. If the price elasticity of demand for the goods is 2 and the original quantity demanded is 30 units, calculate the new quantity demanded. [ISC 2023]

Ans.

$P_1 = ₹100$

$P_2 = ₹80$

$Q_1 = 30$ units

$Q_2 = ?$

$E_d = -2$

$\Delta P = 80 - 100 = -20$

$\Delta Q = Q_2 - 30$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$-2 = \frac{Q_2 - 30}{-20} \times \frac{100}{30}$$

$$-2 = \frac{Q_2 - 30}{-1} \times \frac{1}{6}$$

$12 = Q_2 - 30$

$Q_2 = 42$ units

Hence, the new quantity demanded is 42 units.

Q.3. What is meant by income elasticity of demand? [ISC 2016]

Q.4. If demand increases by 25% due to an increase in income by 50%, calculate the Income Elasticity of Demand. [OEB]

Ans. Income elasticity of demand

$$= \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in Income}}$$

$$= \frac{25}{50} = 0.5$$

Q.5. Complete the following table: [OEB]

E^d	Price (₹)	Quantity demanded
?	50	100
	25	150

Ans.

$P_1 = ₹50$

$P_2 = ₹25$

$Q_1 = 100$ units

$Q_2 = 150$ units

$\Delta P = 25 - 50 = -25$

$\Delta Q = 150 - 100 = 50$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$= \frac{50}{-25} \times \frac{50}{100}$$

$$= \frac{2500}{-2500} = -1$$

3 Marks Questions

Q.1. When price of commodity X changes from ₹40 per unit to ₹20 per unit, its demand increases by 20 units. If price elasticity of demand is 0·5, calculate the initial and the final quantity demand of commodity X. [ISC 2020]

Ans.

$P_1 = 40$

$P_2 = 20$

$\Delta P = 20 - 40 = -20$

$Q_1 = x$

$Q_2 = x + 20$

$\Delta Q = 20$

$e_d = 0.5$

$$e_d = (-)\frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

This question is for practice and its solution is available at the end of the chapter

$$0.5 = (-)\ \frac{20}{-20} \times \frac{40}{x}$$

$$0.5x = 40$$

$$x = 80 = Q_1 \text{ (initial quantity)}$$

$$Q_2 = 80 + 20 = 100 \text{ (final quantity)}$$

[ISC Marking Scheme, 2020]

Examiner's Comment

- Most candidates calculated the initial quantity but did not calculate the final quantity demanded. A common error made by many candidates was that they calculated the final quantity as 60 instead of 100. This error probably occurred because they did not notice that price had fallen in the question and followed the method of subtraction instead of addition of 20 to initial quantity 80.

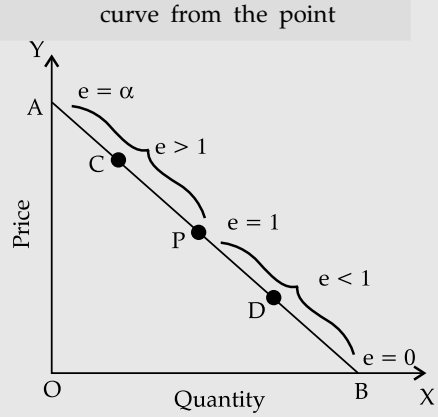
Answering Tip

- In order to solve the numerical on elasticity of demand, students need to first write the formula and then substitute the values in the formula. They need to first calculate the initial quantity and then the final quantity. Emphasis should be on according to the law of demand, if price falls then demand will increase and so change in quantity demanded has to be added to the initial quantity.

Q.2. Explain the various degrees of price elasticity of demand at different points on a straight-line demand curve. [ISC 2019]

Ans. Price elasticity of demand is different at various points on a **downward sloping straight line demand curve**. Price elasticity at various points can be measured by,s

$$E_d = \frac{\text{Lower segment of the demand curve from the point}}{\text{Upper segment of the demand curve from the point}}$$



E_d at point B = $\frac{0}{AB}$ $\therefore E_d = 0$ Perfectly inelastic

E_d at point A = $\frac{AB}{0}$

$\therefore E_d = \propto$ Perfectly elastic PB/AP as PB=AP

E_d at point C = $\frac{CB}{AB}$ as CB > AC

$\therefore E_d > 1$ Relatively elastic

E_d at point M = $\frac{PB}{AP}$ as $\frac{PB}{AP}$

$\therefore E_d = 1$ Unitary elastic

E_d at point D = $\frac{AD}{DB}$ as, DB < AD

$\therefore E_d < 1$ Relatively inelastic

[ISC Marking Scheme, 2019]

Examiner's Comment

- Many candidates explained the degrees of price elasticity of demand through five different diagrams but did not show the degrees of price elasticity of demand on a straight-line demand curve. Several candidates did not write the formula. Some candidates did not join the linear demand curve with the axis and the values of elasticity at x-axis and y-axis were also incorrect. A few candidates, instead of explaining the geometric method, explained the percentage method.

Answering Tips

- Understand Elasticity of Demand with the help of the formula.
- Students need to draw the straight-line demand curve and then to take points at equidistance on the linear demand curve.
- Students need to calculate elasticity of demand at different points by applying the formula.

Q.3. Explain any two factors affecting the price elasticity of demand. [ISC 2018]

Ans. Factors affecting the price elasticity of demand:

- **Nature of Commodity:** If you regard a product as a necessity, then your demand for it will be inelastic: you are willing to pay any reasonable price, e.g. gasoline. If you think it is a luxury, then your demand is very elastic, and it may drop considerably due to an increase in price, e.g. IPL ticket, car.
- **Availability of substitutes:** The more possible substitutes, the greater the elasticity. Example, Coke, Pepsi. If the price of coke goes up, people will be tempted to buy Pepsi. The demand of coke will, therefore, fall. In case of salt, it has no close substitute and is necessary. Its demand is inelastic.
- **Proportion of Total expenditure:** Products that consume a small portion of the consumer's income its demand is inelastic. Example, a consumer

spends a very small proportion of income on purchase of match boxes. Therefore, even large change in its price will not induce him to change his level of demand.

- **Time Period:** Elasticity tends to be greater over the long run because consumers have more time to adjust their behaviour and vice-versa. It is low for a short period.
- **Number of uses:** The greater number of uses of a commodity, the higher is the price elasticity of demand. Example, milk can be used to make cheese, butter, curd, etc. If its price rises, it will be put to only important uses like serving the children or for the sick members in the family.
- **Possibility of postponement:** If the demand for a particular commodity cannot be postponed, its demand will be inelastic, example, medicines, food, etc.
- **Habits:** Those goods which have become habitual necessities for the consumers, have low price elasticity. Example, cigarettes, drugs, etc.
- **Price level:** Highly and low-priced goods have low price elasticity or inelastic demand. Example diamonds and coarse cloths. Medium class commodities are more elastic. Example, watches, cycles, etc. **[Any Two]**

[ISC Marking Scheme, 2018]

Examiner's Comment

- Several candidates, instead of mentioning the effect of the factors on price elasticity wrote about their effect on demand. Many candidates mentioned determinants of demand instead of determinants of price elasticity of demand.

Answering Tips

- Stress that factors affecting price elasticity of demand be written.
- Students should understand that while explaining the factors affecting price elasticity of demand, elastic or inelastic point with the factor needs to be mentioned.

Q.4. Find the elasticity of demand of x and y on the basis of the demand schedule given below and specify which one is more elastic: [ISC 2017]

Good x		Good y	
Px (₹)	Dx (units)	Py (₹)	Dy (units)
8	10	8	10
4	12	6	25

Ans. Elasticity of demand for x good:

Here, $\Delta Q = 2;$
$\Delta P = 4;$
$Q_1 = 10$
$P_1 = 8$

Hence, $E_d = \frac{2}{4} \times \frac{8}{10} = \frac{2}{5} = 0.4$

(inelastic demand or elasticity is less than one)

Elasticity of demand for y good:

Here, $\Delta Q = 15,$
$\Delta P = 2;$
$Q_1 = 10;$
$P_1 = 8$

Hence, $E_d = \frac{15}{2} \times \frac{8}{10} = 6$

(elastic demand or elasticity is more than one)

Thus, the demand for y good is more elastic than the demand for x good. Demand schedule of x good is an example of inelastic demand schedule while the demand schedule of y good is an example of elastic demand schedule.

The demand for good Y is more elastic than the demand for good X. **[ISC Marking Scheme, 2017]**

Examiner's Comment

- In some cases, candidates reached conclusion without calculation of elasticity of demand. Some of the candidates tried to solve it by expenditure method. Several candidates even drew the demand curve using the schedule, which was not correct.

Answering Tip

- Do more practice on calculation of elasticity of demand.

Q.5. The quantity demanded of a commodity is 40 units at a price of ₹10 per unit. Price elasticity of demand is – 2. The price falls by ₹2. What is the new quantity demanded at the new price? A I [ISC 2016]

Ans.
$P_1 = ₹10$
$Q_1 = 40$ units
$Q_2 = ?$
$E_d = -2$
$\Delta P = -2$
$\Delta Q = Q_2 - 40$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$-2 = \frac{Q_2 - 40}{-2} \times \frac{10}{40}$$

$$-2 = \frac{Q_2 - 40}{-1} \times \frac{1}{8}$$

$16 = Q_2 - 40$
$Q_2 = 56$ units

Hence, the new quantity demanded is 56 units

Examiner's Comment

- A number of candidates committed errors in attempting this part. Candidates subtracted 16 instead of adding, to get quantity demanded at new price.

Answering Tip

- Understand the basic logic behind the derivation of the formula of price elasticity and the significance of the negative sign.

Q.6. Draw diagrams to show the elasticity of demand when it is: **[ISC 2015]**

(i) Greater than one (ii) Less than one

(iii) Unity

4 Marks Questions

Q.1. The price elasticity of demand of a goods is (–0.5). At a price of ₹20 per unit its demand is 300 units. At what price will its demand increase by 10%? **[OEB] [CBSE QB]**

Ans.

$E_d = -0.5$

$P_1 = ₹20$

$Q_1 = 300$ units

$P_2 = ?$

$Q_2 = 300 + 10\%$ of $300 = 330$

$\Delta Q = 330 - 300 = 30$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$-0.5 = \frac{-30}{\Delta P} \times \frac{20}{300}$$

$$-0.5 = \frac{1}{\Delta P} \times \frac{2}{1}$$

$-0.5\,\Delta P = 2$

$\Delta P = -4$

$P_2 = P_1 + \Delta P$

$P_2 = ₹20 - ₹4$

$P_2 = ₹16$

Hence, the new price is ₹16

Q.2. The price elasticity of demand of a goods is (–) 1.5. At a price of ₹40 per unit its demand is 300 units. At what price will its demand decrease by 20 percent? **[OEB]**

Q.3. The price elasticity of demand of a goods is – 1. At a price of ₹10 per unit its demand is 500 units. What will be the demand if the price increases by 20 per cent? **[OEB]**

Ans. Given,

$E_d = (-)\,1$

$P_1 = ₹10$

$Q_1 = 500$

$Q_2 = ?$

$P_2 =$ increase by 20%

$P_2 = 20\%$ of $10 + 10 = ₹12$

$\Delta P = 12 - 10 = 2$

$\Delta Q = Q_2 - 500$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$-1 = \frac{Q_2 - 500}{2} \times \frac{10}{500}$$

$$-1 = \frac{Q_2 - 500}{100}$$

$-100 = Q_2 - 500$

$Q_2 = 400$ units.

Q.4. A 5 per cent fall in the price of a goods raises its demand from 300 units to 318 units. Calculate its price elasticity of demand. **[OEB]**

Ans. Given, Percentage change in price = – 5%

$Q_1 = 300$ units

$Q_2 = 318$ units

$\Delta Q = Q_2 - Q_1$

$= 318 - 300$

$= 18$ units

Percentage change in quantity demanded

$$= \frac{18}{300} \times 100 = 6\%$$

Price elasticity of demand (E_d)

$$= \frac{\%\text{ Change in Quantity Demanded}}{\%\text{ Change in Price}}$$

$$= \frac{6}{5}$$

$= -1.2$ (more than unitary elastic) ($e^d > 1$)

6 Marks Questions

Q.1. Explain in detail the methods of measuring price elasticity of demand. **[OEB]**

Ans. Methods for calculating elasticity of Demand:

(i) Percentage Method: In percentage method of price elasticity of demand the coefficient is found with respect to the percentage. It is said to be the degree of percentage change in the quantity demand with the percentage change in the price of the commodity. The percentage method measures price elasticity of demand by dividing the percentage change in amount demanded by percentage change in price of commodity.

(a) The elasticity of demand is unity, greater than unity and less than unity.

(b) Demand is unity if change in demand is proportionate to the change in price.

(c) Demand is greater than unity when change in demand is more than proportionate change in price.

(d) The demand is less than unity if change in demand is less than proportionate change in price.

(e) The coefficient of price elasticity of demand is always negative because change in price brings a change in demand in opposite direction. Negative signs are usually disregarded.

This question is for practice and its solution is available at the end of the chapter

It can be found out using the following formula:

$$\text{Elasticity of Demand} = \frac{\text{Percentage change in the quantity Demanded of the commodity}}{\text{Percentage Change in the Price of the commodity}}$$

OR

$$E_d = \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100}$$

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

(ii) Arc Elasticity Method: Arc elasticity of demand measures elasticity between two points on a curve – using a mid-point between the two curves.

On most curves, the elasticity of a curve varies depending on where you are. Therefore elasticity needs to measure a certain sector of the curve.

Calculating Arc Elasticity of Demand

To calculate arc elasticity of demand we first take the midpoint in between.

$$\text{Midpoint } Q = \frac{(Q1 + Q2)}{2}$$

$$\text{Midpoint } P = \frac{(P1 + P2)}{2}$$

Once we have the midpoint, we calculate the PED in the usual way

Arc elasticity of demand (PED)

$$= \frac{Q2 - Q1 / \text{midpoint } Q}{P2 - P1 / \text{midpoint } P}$$

(iii) Total Expenditure Method: Under this method, impact (effect) of change in price on the expenditure of a goods is studied. When price of a goods changes, consumer's total expenditure on it may increase, decrease, or remain constant. Thus, elasticity is measured by comparing the total expenditure made on the goods before and after the price change.

(a) If total expenditure on a commodity remains unchanged before and after the price change, the elasticity is said to be unity $E_d = 1$.

(b) If total expenditure increase with fall in price (and vice-versa), elasticity of demand is said to be greater than unity $E_d > 1$.

(c) If total expenditure decreases with fall in price (and vice-versa), elasticity of demand is said to be less than unity $E_d < 1$.

(iv) Geometric or Point Method: Geometric method was suggested by **Prof. Marshall** and is used to measure the elasticity at a point on the demand curve. When there are infinitely small changes in price and demand, then the 'Geometric Method' is used. This method is also known as 'Graphic Method' or 'Point Method' or 'Arc Method'. Elasticity of demand (Ed) is different at different points on the same straight line demand curve.

In order to measure Ed at any particular point, lower portion of the curve from that point is divided by the upper portion of the curve from the same point.

Formula: Elasticity of Demand (Ed)

$$= \frac{\text{Lower segment of demand curve (LS)}}{\text{Upper segment of demand curve (US)}}$$

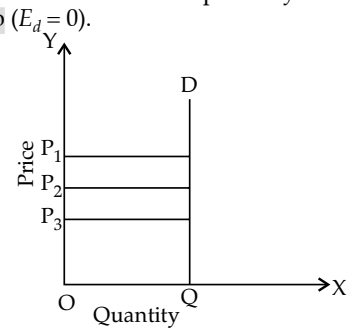
[Any Two]

Q.2. "Elasticity of demand depends on a large number of factors." In the light of this statement explain the various factors on which the elasticity of demand is affected. A I **[OEB]**

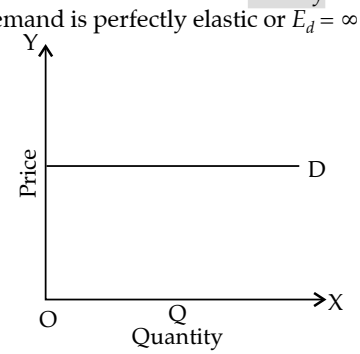
Q.3. There are a lot of degrees of price elasticity of demand. Justify. [OEB]

Ans. Types/Degrees of Elasticity:

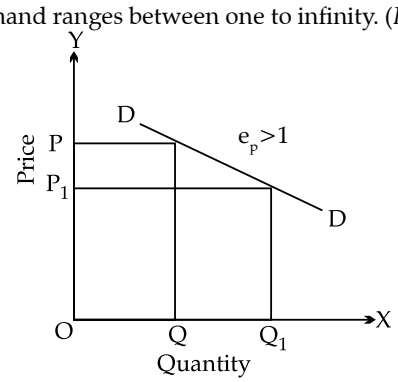
(i) Perfectly Inelastic Demand: A perfectly inelastic demand is one when there is no change produced in the demand of a product with change in its price. The numerical value for perfectly inelastic demand is zero ($E_d = 0$).



(ii) Perfectly Elastic Demand: When a small change in price of a product causes a major change in its demand, it is said to be perfectly elastic demand. In perfectly elastic demand, a small rise in price results in fall in demand to zero, while a small fall in price causes increase in demand to infinity. In such a case, the demand is perfectly elastic or $E_d = \infty$

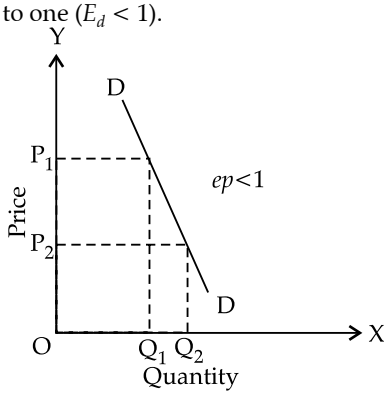


(iii) Relatively Elastic Demand: Relatively elastic demand refers to the demand when the proportionate change produced in demand is greater than the proportionate change in price of a product. The numerical value of relatively elastic demand ranges between one to infinity. ($E_d > 1$)

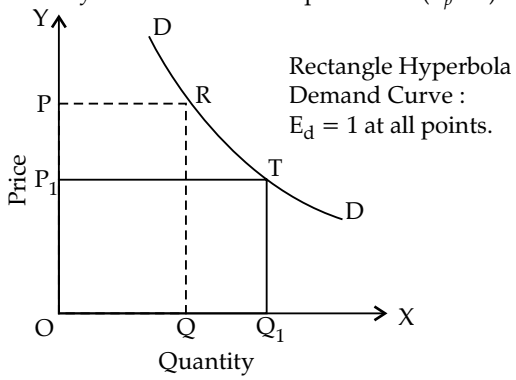


This question is for practice and its solution is available at the end of the chapter

(iv) Relatively Inelastic Demand: Relatively inelastic demand is one when the percentage change produced in demand is less than the percentage change in the price of a product. The numerical value of relatively elastic demand ranges between zero to one ($E_d < 1$).



(v) Unitary Elastic Demand: When the proportionate change in demand produces the same change in the price of the product, the demand is referred as unitary elastic demand. The numerical value for unitary elastic demand is equal to one ($E_p = 1$).



OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. If the price of a commodity and total expenditure on that commodity change in the same direction, the price elasticity of demand will be:

[ISC 2023]

(A) greater than one **(B)** less than one
(C) equal to zero **(D)** equal to infinity

Ans. Option (B) is correct.

Explanation: When price and total expenditure of a commodity increases in same direction, it means nature of price elasticity of commodity is less elastic demand. It occurs when the change in the amount of a product demanded in due proportion is less than the change in price in due proportion.

Q.2. A demand curve parallel to X axis signifies:

[ISC Specimen Paper, 2022-23]

(A) perfectly elastic
(B) perfectly inelastic
(C) elastic
(D) inelastic

Q.3. If two goods are complementary, then their:

[ISC Semester-I, 2022]

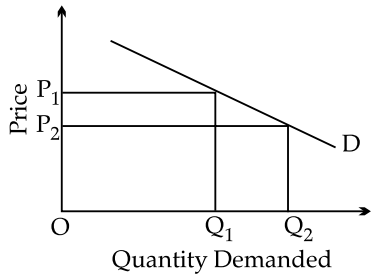
(A) Income elasticity are negative.
(B) Cross price elasticity is positive.
(C) Cross price elasticity is negative.
(D) Cross price elasticity is zero.

Ans. Option (C) is correct.

Explanation: As there is an inverse relationship, that is when the price of the complement rises. The demand for the good falls, the cross elasticity of complementary goods will always be negative.

Q.4. In the diagram above, the demand curve represents:

[ISC Semester-I, 2022]



(A) Unit elastic **(B)** Perfect inelastic
(C) Highly elastic **(D)** Highly inelastic

Ans. Option (C) is correct.

Explanation: When we see in the figure, the change in quantity demanded is more than the change in its price, thus the elasticity of demand in this case will be highly elastic.

Q.5. At the midpoint of a downward sloping straight demand curve touching two axis:

[ISC Semester-I, 2022]

(A) $E_d = 2$ **(B)** $E_d = 0.5$ **(C)** $E_d = 1$ **(D)** $E_d = 4$

Ans. Option (C) is correct.

Explanation: At midpoint, as the point divides the line into two halves, the elasticity will be unitary and equal to one as both the lower segment and upper segment will be equal.

Q.6. When Zubair had a hike in his income, he switched over to rice and wheat from consumption of maize and jowar. The income elasticity for the latter goods is: **[ISC Semester-I, 2022]**

(A) Positive **(B)** Negative
(C) Zero **(D)** Constant

Ans. Option (B) is correct.

Explanation: In the case of inferior goods, the income elasticity of demand is negative. That is, when the income of the consumer rises, the demand for inferior goods fall and when the income of the consumer falls, then the demand for inferior goods rises.

This question is for practice and its solution is available at the end of the chapter

Q.7. Which of the given options shows the effects of Elasticity of demand?
[ISC Specimen Paper, Semester-I, 2022]
(A) Nature of good
(B) Level of Income
(C) Availability of substitutes
(D) All of the above

Q.8. The demand for a certain brand of fruit juice is elastic because:
[ISC Specimen Paper, Semester-I, 2022]
(A) It falls under luxuries.
(B) Its consumption can be postponed.
(C) Its substitutes are available in the market.
(D) It is a multiple use product.

Ans. Option (C) is correct.
Explanation: For a fruit juice brand there are a lot of substitutes available, making the elasticity of demand elastic.

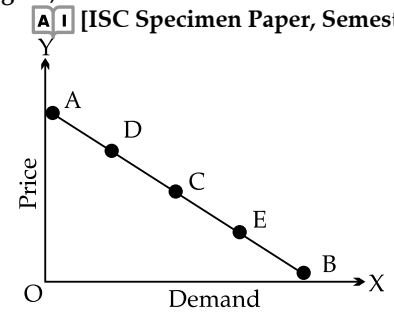
Q.9. Cross elasticity of demand is _____ for substitute goods. An increase in the price of one commodity leads to _____ in the demand for another commodity.
[ISC Specimen Paper, Semester-I, 2022]
(A) positive, increase
(B) negative, increase
(C) positive, decrease
(D) negative, increase

Ans. Option (A) is correct.

Q.10. When a fall in the price of the commodity results in increase in total expenditure, the elasticity of demand will be _____.
[ISC Specimen Paper, Semester-I, 2022]
(A) $E_p = 1$ **(B)** $E_p = 0$
(C) $E_p < 1$ **(D)** $E_p > 1$

Ans. Option (D) is correct.
Explanation: When there is a fall in the price but still the total expenditure is increasing, means that the increase in the quantity demanded is more than the decrease in the price of the commodity. Thus the elasticity in such a case would be greater than unity.

Q.11. In figure, the demand is:
[ISC Specimen Paper, Semester-I, 2022]



(A) Perfectly elastic at A, unit elastic at B and relatively inelastic at E.
(B) Relatively elastic at D, perfectly elastic at C and unit elastic at E.
(C) Unit elastic at C, relatively inelastic at E and perfectly inelastic at B.
(D) Relatively inelastic at D, unit elastic at C and perfectly elastic at B.

Ans. Option (C) is correct.
Explanation: At point C, the lower segment is equal to upper segment, making it unitary elastic. At point E, the upper segment is more than the lower segment making it relatively inelastic. Point B is on the X – axis, thus the point shows perfectly elastic demand.

Q.12. A consumer buys 80 units of a good at a price of ₹8 per unit. Price falls to ₹6 per unit. How much quantity will the consumer buy at a new price if ep = (–) 2? **[ISC Specimen Paper, Semester–I, 2022]**
(A) 100 units **(B)** 110 units
(C) 120 units **(D)** 130 units.

Ans. Option (C) is correct.
Explanation:
$P_1 = 8$
$P_2 = 6$
$\Delta P = 6 - 8 = -2$
$Q_1 = 80$
$Q_2 = x$
$\Delta Q = x - 80$
$ed = -2$
$$ed = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$
$$-2 = \frac{x-80}{-2} \times \frac{8}{80}$$
$40 = x - 80$
$x = 120$ units (final quantity)

Q.13. Due to the pandemic the prices of vegetables and fruits have increased but the demand for quantity has not fallen.
What is the reason for the demand remaining the same? **[ISC Specimen Paper, Semester-I, 2022]**
(A) Demand for food is inelastic since food is a necessity.
(B) Demand for food is elastic
(C) Demand is for the inferior goods.
(D) Demand is for the luxury goods.

Ans. Option (A) is correct.
Explanation: For necessaries of life the demand is inelastic because people buy the required amount of goods whatever their price. For example, necessaries such as rice, salt, cloth etc. are purchased whether they are expensive or cheap.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): Demand for salt is inelastic.
Reason (R): In case of elastic demand, percentage change in price of a commodity causes relatively less than percentage change in quantity demanded.
[OEB]

Ans. Option (C) is correct.
Explanation: Reason (R) is false as when the percentage change in the price of a commodity is less than the percentage change in quantity demanded, it will result in less than unity elasticity, that is inelastic demand.

This question is for practice and its solution is available at the end of the chapter

Q.2. Assertion (A): Demand for antiques is perfectly inelastic.
Reason (R): The perfectly inelastic demand has the elasticity equal to zero. **[OEB]**

Ans. Option (B) is correct.
Explanation: Demand for antiques is perfectly inelastic not because the elasticity is equal to zero but because the quantity demanded will not change even if the price changes.

Q.3. Assertion (A): Demand for refrigerator is unitary elastic.
Reason (R): The unitary elastic demand has elasticity equal to one. **[OEB]**

Q.4. Assertion (A): The demand for milk is relatively more elastic than the demand for gold.
Reason (R): Milk can be used for multiple purposes, but that of gold is not. **[OEB]**

Ans. Option (D) is correct.
Explanation: The demand for milk is relatively inelastic because it is a necessity.

Q.5. Assertion (A): Cross elasticity of demand is negative for complements.
Reason (R): Complements are the goods which are used together and so when the price of such goods rises, the demand for its complements also falls as they become relatively more expensive to use. **[OEB]**

Ans. Option (A) is correct.
Explanation: As complement goods so an inverse relationship, their cross elasticity is negative.

Case Based Questions

[A] Multiple Choice Questions

I. Answer the questions with respect to the case: **[ISC Semester-I, 2022]**

When Price of a good is ₹12 per unit, 24 units are demanded. When the price rises to ₹14 per unit, 20 units are demanded.

(a) The price elasticity is:
(A) One
(B) Greater than one
(C) Zero
(D) Between zero and one.

Ans. Option (A) is correct.
Explanation: $P_1 = ₹12; P_2 = ₹14; Q_1 = 24, Q_2 = 20$

$$\Delta P = 14 - 12 = 2$$
$$\Delta Q = 20 - 24 = -4$$
$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$
$$= \frac{-4}{2} \times \frac{12}{24} = -1$$

(b) The shape of the above demand curve will be:
(A) Positively sloped. **(B)** Negatively sloped.
(C) Parallel to X axis. **(D)** Parallel to Y axis.

Ans. Option (B) is correct.
Explanation: In the numerical question itself it can be seen that there is an inverse relationship between the price and the quantity. Thus, the demand curve will be a rectangular hyperbola and negatively sloped.

(c) The price elasticity of demand measures:
(A) Slope of budget line.
(B) Percentage change in the price of the good.
(C) Responsiveness of quantity demanded to the change in price.
(D) Sensitiveness of price to change in quantity demanded.

Ans. Option (C) is correct.

[B] Subjective Based Questions

II. Rajiv and Smita and two different people living in the same society. Rajiv is a chain smoker and so requires buying cigarette even if the prices is very high. Smita on the other hand is addicted to tea, so she needs to drink tea every moment. Looking at the demand pattern, she can't sustain to drink coffee its substitute. Smita uses milk for making a lot of items, like butter, curd, ghee, sweets and lot more. The price of milk increased and so she started using it for the most important items and not for all of the things that she makes.
Talking about Rajiv, he has been wanting to buy a new car since a long time, but due to other responsibilities, he has been postponing it. **[OEB]**

(a) What is the elasticity of demand for cigarette for Rajiv and tea for Smita?

Ans. The elasticity of demand for cigarette for Rajiv and tea for Smita is inelastic.

(b) Why does Rajiv and Smita face such an elasticity of demand?

Ans. Rajiv and Smita face such an elasticity of demand as the cigarette for Rajiv and tea for Smita is a part of their habit and they cannot do without it even if their prices rise.

(c) Explain the factors affecting the elasticity of demand as stated in the above passage.

Ans. The factors affecting the elasticity of demand are:
(i) Alternative Uses of a Commodity: The greater number of uses of a commodity, the higher is the price elasticity of demand. Example, milk can be used to make cheese, butter, curd, etc. If its price rises, it will be put to only important uses like serving the children or for the sick members in the family.
(ii) Postponement of Use: Usually the demand for such commodities whose use can be postponed for some time is elastic. For example: the demand for car is elastic because its use can be postponed for some time, if its price goes up.

(d) Even though tea and coffee are substitutes, why can't it be the case above?

Ans. Even though tea and coffee are substitutes, it is not as such in the above case for Smita as she prefers to have tea and not coffee. So, in this case coffee is not considered as a substitute for tea, making the demand for tea inelastic.

(e) Explain the factor nature of the commodity which affects elasticity of demand and is not mentioned in the above passage.

Ans. Generally, all commodities can be divided into three categories are:
(i) Necessities of Life: For necessities of life the demand is inelastic because people buy the required amount of goods whatever their price. For example, necessaries such as rice, salt, cloth etc. are purchased whether they are expensive or cheap.
(ii) Conventional Necessities: The demand for conventional necessities is less elastic or inelastic.

This question is for practice and its solution is available at the end of the chapter

People are accustomed to the use of goods like intoxicants which they purchase at any price. For example, drunkards consider opium and wine almost as a necessity as food and water. Therefore, they buy the same amount even when their prices are higher.

(iii) Luxury Commodities: The demand for luxury is usually elastic as people buy more of them at a lower price and less at a higher price. For example, the demand of luxuries like silk, perfumes and ornaments increases at a lower price and decreases at a higher price.

III. Read the extract given below and answer the questions that follows: [OEB]

In this method of price elasticity of demand the coefficient is found with respect to the percentage. It is said to be the degree of percentage change in the quantity demand with the percentage change in the price of the commodity.

The method measures price elasticity of demand by dividing the percentage change in amount demanded by percentage change in price of commodity.

(i) The elasticity of demand is unity, greater than unity and less than unity.

(ii) Demand is unity if change in demand is proportionate to the change in price.

(iii) Demand is greater than unity when change in demand is more than proportionate change in price.

(iv) The demand is less than unity if change in demand is less than proportionate change in price.

(v) The coefficient of price elasticity of demand is always negative because change in price brings a change in demand in opposite direction. Negative signs are usually disregarded.

It can be found out using the following formula:

Price Elasticity of demand, E_d

$$= \frac{\text{Percentage change in the quantity demanded of the commodity}}{\text{Percentage change in the price of the commodity}}$$

(a) Which method of price elasticity is the above passage talking about?

Ans. Percentage method

(b) What are the other methods that can be used to measure the elasticity of demand?

Ans. The other methods are:

(i) Point Method or Geometric Method.

(ii) Total Expenditure Method.

(iii) Arc Elasticity Method.

(c) A consumer purchased 100 units of a good at ₹15. When the price decreased to ₹10, the demand increased by 50. Calculate the price elasticity of demand using the method as explained in the passage.

Ans. Given, $Q = 100$

$P = ₹15$

Change in Quantity, $\Delta Q = 50$

Change in Price, $\Delta P = ₹15 - ₹10 = ₹5$

We know that:

Elasticity of Demand, $e_d = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$

$\Rightarrow \quad e_d = \frac{50}{5} \times \frac{15}{100}$

$\therefore \quad e_d = 1.5$

(d) When the price of the commodity say apples was ₹100, the quantity demanded was 2 kg. When the price falls to ₹80 the quantity demanded rises to 3 kg of apples. In such a case find the price elasticity using the above method.

Ans. $\Delta Q = 3 - 2 = 1$ kg

$\Delta P = 100 - 80 = 20$

$Q = 2$ kg

$P = 100$

Using the formula $E_d = \frac{1}{20} \times \frac{100}{2}$

The elasticity of demand will be = 2.5 in this case

(e) Differentiate between income elasticity and cross elasticity of demand.

Ans. The difference between income elasticity of demand and cross elasticity of demand are:

S. No.	Basis of Distinction	Income Elasticity of Demand	Cross Elasticity of Demand
1.	Meaning	The responsiveness of change in the quantity demanded of a commodity with respect to the change in the income of the consumer.	The responsiveness of the change in the quantity demanded of a commodity with respect to the change in the price of a related commodity.
2.	Formula	$E_d = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$	$E_d = \frac{\Delta Q_x}{\Delta P_Y} \times \frac{P_Y}{Q_x}$

Solutions for Practice Questions

2 Marks Questions

3. Income elasticity of demand: It is the measure/ degree of responsiveness of quantity demanded of any commodity to the change in income of the consumers.

OR

$$\text{Income elasticity} = \frac{\%\,\Delta \text{ in quantity demanded}}{\%\,\Delta \text{ in income of the consumer}}$$

[ISC Marking Scheme, 2016]

Examiner's Comment

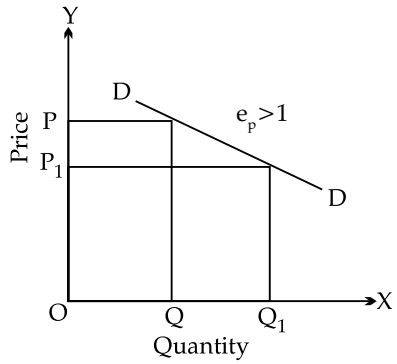
- Some of the candidates did not mention 'degree of responsiveness' of demand to change in income. Several candidates wrote 'price' instead of 'income'.

Answering Tip

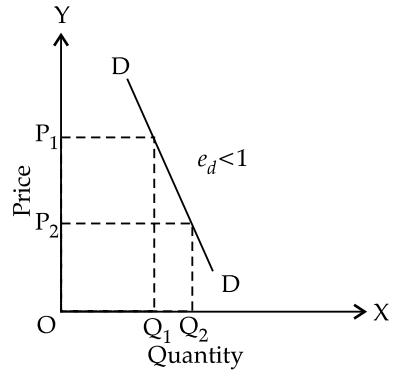
- Emphasise the importance of the 'degree of responsiveness' in the definition of elasticity.

3 Marks Questions

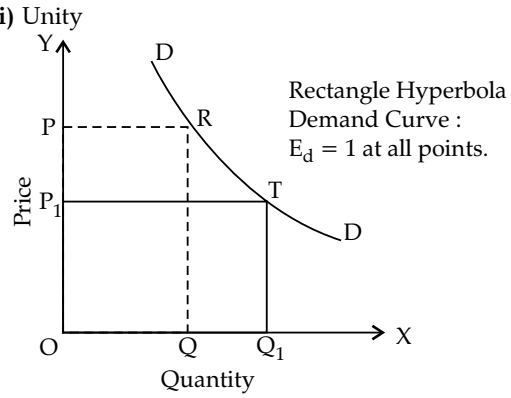
6. (i) Greater than one



(ii) Less than one



(iii) Unity



Examiner's Comment

- In several cases, the demand curve drawn was steeper instead of it being flatter. Proportionate change in price and demand was not shown correctly. Some candidates did not draw a rectangular hyperbola to show unity. The three diagrams drawn did not show proper variations. In some cases, $E_d>1$, $E_d<1$ and $E_d = 1$ were shown on the same demand curve.

Answering Tip

- Learn the degrees of elasticity of demand properly. Proper diagram showing $E_d>1$, $E_d<1$ and $E_d = 1$ should be taught.

4 Marks Questions

2. Given,

$$E_d = (-)\ 0.5$$
$$P_1 = ₹40$$
$$Q_1 = 300$$
$$P_2 = ?$$
$$Q_2 = 300 - 20\% \text{ of } 300 = 240$$
$$\Delta P = P_2 - 40$$
$$\Delta Q = 300 - 240 = -60$$
$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$
$$-1.5 = \frac{-60}{P_2 - 40} \times \frac{40}{300}$$
$$1.5 = \frac{8}{P_2 - 40}$$
$$1.5\,P_2 - 60 = 8$$
$$1.5\,P_2 = 68$$
$$P_2 = 45.33$$

Hence, the new price is ₹45.33

6 Marks Questions

2. Factors Determining the Elasticity of Demand:

(i) Nature of commodity: Generally, all commodities can be divided into three categories are:

(a) Necessaries of Life: For necessaries of life the demand is inelastic because people buy the required amount of goods whatever their price. For example, necessaries such as rice, salt, cloth etc. are purchased whether they are expensive or cheap.

(b) Conventional Necessities: The demand for conventional necessities is less elastic or inelastic. People are accustomed to the use of goods like intoxicants which they purchase at any price. For example, drunkards consider opium and wine almost as a necessity as food and water. Therefore, they buy the same amount even when their prices are higher.

(c) **Luxury Commodities:** The demand for luxury is usually elastic as people buy more of them at a lower price and less at a higher price. For example, the demand of luxuries like silk, perfumes and ornaments increases at a lower price and decreases at a higher price.

(ii) **Existence of substitutes:** Demand for goods which have close substitutes (like tea and coffee) is relatively more elastic because, when price of such good rises, the consumers have the option of shifting to its substitute. Goods without close substitutes like cigarettes, etc., are generally found to be less elastic in demand.

(iii) **Portion of income spent on a commodity:** Demand for goods on which a consumer spends only a small fraction of his total income is relatively more inelastic because, even when price of such good rises, the consumers can still afford the same units of the commodity. Goods on which a major portion of the income is spent are likely to have elastic demand.

(iv) **Alternative Uses of a Commodity:** The greater number of uses of a commodity, the higher is the price elasticity of demand. Example, milk can be used to make cheese, butter, curd, etc. If its price rises, it will be put to only important uses like serving the children or for the sick members in the family.

(v) **Price Level:** Higher the level of price, higher is the elasticity of demand for a commodity.

(vi) **Postponement of Use:** Usually the demand for such commodities whose use can be postponed for some time is elastic. For example: the demand for DVD player is elastic because its use can be postponed for some time, if its price goes up, but the demand for wheat and rice is inelastic because their use cannot be postponed even when price goes up.

(vii) **Time Period:** Elasticity of demand is high over a long period (compared to a short period), because during a short period of time, consumption habits tend to be stable

(viii) **Tastes and Habits of the consumer:** Demand for goods for which a consumer has tastes preference or is habitual of it is relatively more inelastic because, even when price of such good rises, the consumers will still prefer to buy the same units of the commodity. **[Any Six]**

Solutions for Objective Type Questions

[A] Multiple Choice Questions

2. Option (A) is correct.

Explanation: The demand curve parallel to X axis indicates that even when there is no change in the price of the commodity the demand changes infinitely. Hence the elasticity of demand is infinity.

7. Option (D) is correct.

Explanation: Nature of commodity: Generally, all commodities can be divided into three categories are:

(i) **Necessaries of Life:** For necessaries of life the demand is inelastic because people buy the required amount of goods whatever their price. For example, necessaries such as rice, salt, cloth etc. are purchased whether they are expensive or cheap.

(ii) **Conventional Necessities:** The demand for conventional necessities is less elastic or inelastic. People are accustomed to the use of goods like intoxicants which they purchase at any price. For example, drunkards consider opium and wine almost as a necessity as food and water. Therefore, they buy the same amount even when their prices are higher.

(iii) **Luxury Commodities:** The demand for luxury is usually elastic as people buy more of them at a lower price and less at a higher price. For example, the demand of luxuries like silk, perfumes and ornaments increases at a lower price and decreases at a higher price.

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Level of income: Demand for goods on which a consumer spends only a small fraction of his total income is relatively more inelastic because, even when price of such good rises, the consumers can still afford the same units of the commodity. Goods on which a major portion of the income is spent are likely to have elastic demand.

[B] Assertion Reason Questions

3. Option (D) is correct.

Explanation: Demand for refrigerator is highly elastic as the demand will change drastically with the change in its price. This makes assertion false. Reason is true as the unitary elastic demand has elasticity equal to one.

REFLECTION

(a) Do you understand the meaning of elasticity and various types of elasticity?

(b) Can you explain the factors which affect the elasticity of demand?

(c) Can you show with diagrams, the various measures of elasticity of demand?

(d) Will you be able to calculate the price elasticity of demand using various methods?